

SUM OF INTEGERS FROM 1 TO N USING PROLOG

Aim

To write and execute a Prolog program to find the sum of all integers from **1 to N** using recursion.

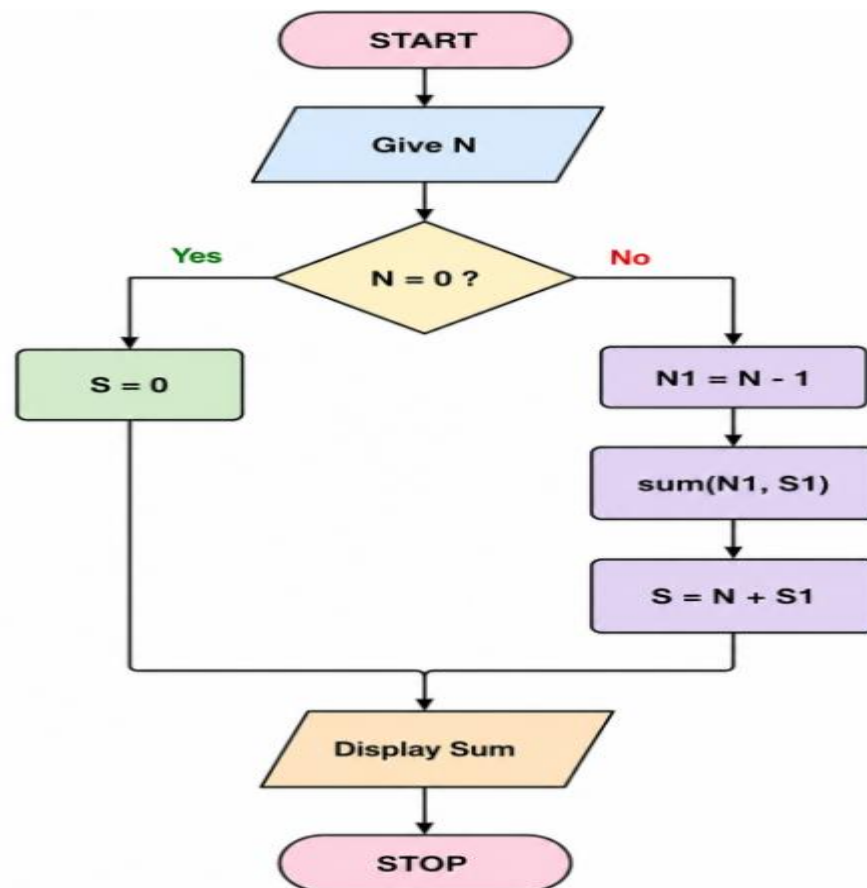
Objectives

- To understand the basic concepts of Prolog programming.
- To understand recursion in Prolog.
- To calculate the sum of integers from 1 to a given number.
- To understand facts, rules, and queries in Prolog.
- To execute the program using SWI-Prolog.

Algorithm

1. Start.
2. Read the value of N.
3. Define the base condition: if $N = 0$, the sum is 0.
4. Check whether N is greater than 0.
5. Calculate $N1 = N - 1$.
6. Recursively find the sum of integers from 1 to N1.
7. Add N to the previous sum.
8. Store the result in S.
9. Display the final sum.
10. Stop.

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Program

```

athin.pl
File Edit Browse Compile Prolog Pce Help
athin.pl
sum(0, 0) .

sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is N + S1.

```

OUTPUT

 SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)

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Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run `?- license.` for legal details.

For online help and background, visit <https://www.swi-prolog.org>
For built-in help, use `?- help(Topic).` or `?- apropos(Word).`

`?- sum(5, S).`
`S = 15 ,`

`?- sum(100, S).`
`S = 5050 ,`

Result

The Prolog program was successfully implemented and executed to calculate the **sum of integers from 1 to N using recursion**. For the input $N = 5$, the obtained sum is **15**.